DAY 14 – PORTS AND PROTOCOLS

1. INTRODUCTION

In computer networking, ports and protocols play an important role in enabling communication between devices and applications.

A port is a logical communication endpoint identified by a number, while a protocol is a set of rules that defines how data is transmitted and received across a network.

Different network services use specific port numbers and protocols. For example, HTTP commonly uses port 80, HTTPS uses port 443, and DNS uses port 53.

Understanding ports and protocols is essential in cybersecurity because it helps identify network services, monitor traffic, configure firewalls, and detect suspicious network activity.

2. OBJECTIVE

The main objective of this task is to understand commonly used network ports, protocols, and their functions.

Specific Objectives

1. Understand the concept of network ports.
2. Identify commonly used port numbers.
3. Learn the purpose of network protocols.
4. Understand the difference between TCP and UDP.
5. Create a port reference sheet.
6. Understand the importance of ports in cybersecurity.

3. WHAT IS A NETWORK PORT?

A network port is a logical endpoint used by transport protocols such as TCP and UDP to identify specific applications or services running on a device.

Port numbers range from 0 to 65535.

Classification of Ports

Port Range	Category	Description
0–1023	Well-Known Ports	Commonly assigned to standard network services
1024–49151	Registered Ports	Assigned or registered for applications and services
49152–65535	Dynamic/Private Ports	Commonly used for temporary client connections

4. WHAT IS A NETWORK PROTOCOL?

A network protocol is a set of rules that controls communication between devices over a network.

Protocols define how information is formatted, transmitted, received, and interpreted.

Examples

- HTTP – Web communication
- HTTPS – Secure web communication
- DNS – Domain name resolution
- SSH – Secure remote access
- SMTP – Email transmission
- FTP – File transfer
- DHCP – Automatic IP address configuration

5. COMMON NETWORK PORTS AND PROTOCOLS

The following table provides a reference for commonly used network services.

Port Number	Protocol/Service	Transport	Purpose
20	FTP Data	TCP	Traditional FTP data transfer
21	FTP Control	TCP	File transfer control
22	SSH	TCP	Secure remote login
23	Telnet	TCP	Unencrypted remote access
25	SMTP	TCP	Email transmission between servers
53	DNS	TCP/UDP	Domain name resolution
67	DHCP Server	UDP	Automatic IP configuration
68	DHCP Client	UDP	Receives DHCP configuration
69	TFTP	UDP	Simple file transfer
80	HTTP	TCP	Web communication
110	POP3	TCP	Email retrieval
123	NTP	UDP	Time synchronization
143	IMAP	TCP	Email access
161	SNMP	UDP	Network monitoring

162	SNMP Traps	UDP	Network notifications
389	LDAP	TCP/UDP	Directory services
443	HTTPS	TCP	Secure web communication
445	SMB	TCP	File and printer sharing
465	SMTP Submission (TLS)	TCP	Encrypted email submission
587	SMTP Submission	TCP	Email submission
636	LDAPS	TCP	LDAP over TLS
993	IMAPS	TCP	Secure IMAP
995	POP3S	TCP	Secure POP3
1433	Microsoft SQL Server	TCP	Database communication
3306	MySQL	TCP	Database communication
3389	RDP	TCP/UDP	Remote desktop access
5432	PostgreSQL	TCP	Database communication
8080	HTTP Alternate	TCP	Alternative web service port

6. IMPORTANT NETWORK PROTOCOLS

6.1 HTTP (Hypertext Transfer Protocol)

HTTP is an application-layer protocol used for transferring web pages and other web resources.

Default Port: 80 (TCP)

Purpose: Communication between web browsers and web servers.

Security: HTTP does not provide encryption by itself.

6.2 HTTPS (Hypertext Transfer Protocol Secure)

HTTPS is HTTP protected using TLS encryption.

Default Port: 443 (TCP), or UDP 443 for HTTP/3.

Purpose: Secure communication between web browsers and websites.

Security: Protects the confidentiality and integrity of data during transmission and authenticates the server when certificates are properly validated.

6.3 DNS (Domain Name System)

DNS translates domain names into IP addresses and provides other naming information.

Port: 53 (UDP/TCP)

Example: Converting www.google.com into an IP address.

Purpose: Helps devices locate websites and network services.

6.4 SSH (Secure Shell)

SSH is a network protocol used for secure remote access and administration.

Default Port: 22 (TCP)

Purpose: Securely access and manage remote computers and servers.

Security: Uses encryption and authentication to protect communication.

6.5 SMTP (Simple Mail Transfer Protocol)

SMTP is used for sending and transferring email messages.

Common Ports: 25, 587, and 465 (TCP)

Purpose: Email transmission between servers and email submission by clients.

6.6 FTP (File Transfer Protocol)

FTP is used to transfer files between computers.

Common Ports: 21 for control and 20 for traditional active-mode data transfer.

Security: Traditional FTP does not encrypt credentials or transferred data.

6.7 DHCP (Dynamic Host Configuration Protocol)

DHCP automatically provides IP addresses and other network configuration details to devices.

Ports: UDP 67 and 68 for DHCPv4.

Purpose: Simplifies network configuration.

7. DIFFERENCE BETWEEN TCP AND UDP

TCP	UDP
Connection-oriented	Connectionless
Provides reliable delivery	Does not guarantee delivery
Maintains data ordering	Does not guarantee ordering
Uses acknowledgements and retransmissions	No built-in acknowledgements or retransmissions

Used by SSH and traditional HTTP/HTTPS	Used by DNS queries, DHCP, and NTP

8. HTTP VS HTTPS

HTTP	HTTPS
Commonly uses TCP port 80	Commonly uses port 443
Does not provide encryption itself	Uses TLS encryption
Data can be intercepted or modified more easily	Protects data in transit
Suitable only where transport security is unnecessary	Preferred for secure web communication

9. IMPORTANCE OF PORTS IN CYBERSECURITY

Understanding network ports helps cybersecurity professionals identify active services and detect potential security risks.

1. Network Monitoring

Monitoring port activity helps identify unusual communication patterns.

2. Firewall Configuration

Firewalls can allow or block network traffic based on IP addresses, protocols, and port numbers.

3. Vulnerability Identification

Unnecessary open ports may expose services to attackers.

4. Incident Investigation

Security teams examine network connections and port activity during incident investigations.

5. Secure Network Configuration

Disabling unnecessary services and restricting access to sensitive ports reduces the attack surface.

10. COMMON PORT-RELATED SECURITY RISKS

Port	Service	Potential Security Risk
21	FTP	Unencrypted credentials and data
23	Telnet	Unencrypted remote access
22	SSH	Unauthorized access if poorly secured
25	SMTP	Email server abuse or misconfiguration

53	DNS	DNS spoofing or amplification attacks
80	HTTP	Unencrypted communication
445	SMB	Unauthorized file access or exploitation
3389	RDP	Unauthorized remote desktop access
3306	MySQL	Unauthorized database access if exposed

11. PORT SECURITY BEST PRACTICES

1. Close unnecessary ports and disable unused services.
2. Configure firewall rules properly.
3. Use secure protocols such as HTTPS and SSH.
4. Avoid insecure protocols such as Telnet.
5. Keep network services updated.
6. Restrict remote administration access.
7. Monitor suspicious network connections.
8. Use strong authentication and MFA where supported.
9. Review exposed services regularly.
10. Follow the principle of least privilege.

13. CONCLUSION

Ports and protocols are essential components of computer networking. They enable devices and applications to communicate using standardized methods.

Common protocols such as HTTP, HTTPS, DNS, SSH, SMTP, FTP, and DHCP perform different networking functions and use commonly assigned port numbers.

Understanding ports and protocols helps cybersecurity professionals monitor network activity, identify exposed services, configure firewalls, and reduce security risks.

Maintaining secure network configurations, restricting unnecessary ports, and using encrypted communication protocols are important practices for protecting organizational networks.